

AP Precalculus Unit 1

Polynomial and Rational Functions

Complete Practice Workbook

Purpose

A rigorous student workbook containing 168 constructed-response problems, 56 AP-style multiple-choice questions, and 28 six-point free-response tasks.

ECHS Mathematics

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Strengthened 2026 classroom edition

Topic 1.1 - Change in Tandem

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe how input and output values vary together, including sign and direction on intervals.
- Connect verbal, numerical, graphical, and symbolic representations of a function.
- Construct and critique contextual graphs with justified axes, units, anchors, domain, range, and phases.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Determine whether the relation is a function of x .

x	-3	-1	0	2	2	5
y	4	4	1	-2	3	-2

Question 2

Representation

No calculator

A function has pairs $(-4, 6)$, $(-1, 0)$, $(2, -3)$, $(7, 0)$. State the displayed domain and range, find $f(2)$, and solve $f(x) = 0$ from the display.

Section A continued

Question 3

Reasoning

No calculator

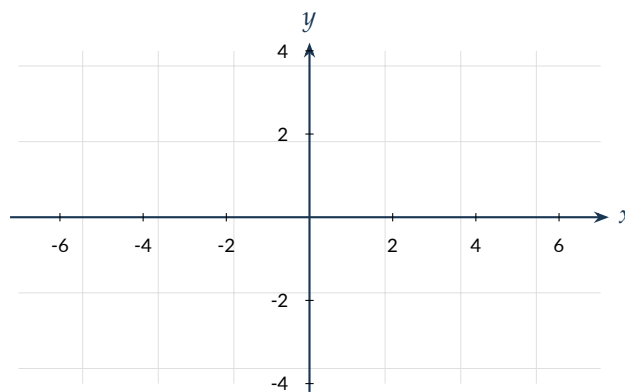
On $1 \leq x \leq 5$, a function is positive and decreasing. Write a complete inequality comparing $f(1)$, $f(3)$, $f(5)$, and 0.

Question 4

Modeling

No calculator

A cyclist is 0 km from school at 7:00, reaches 6 km at 7:20, stays there until 7:30, and returns by 7:42. Draw a labeled graph and identify the segment with greatest rate magnitude.



Question 5

Connections

No calculator

For $f(x) = x^2 - 4$ and $g(x) = |x| - 2$, compare their signs at $x = 1$ and $x = 3$. Why does agreement in sign not make the functions identical?

Section B: Application and reasoning**Question 6**

Algebra

No calculator

If $h(x) = 3x + k$ and $h(-2) = 11$, find k , the preimage of 2, and the direction of h .

Question 7

Piecewise

No calculator

Let $p(x) = 2x + 5$ for $x < 1$ and $p(x) = 10 - x$ for $x \geq 1$. Find $p(-3)$, $p(4)$, and every solution of $p(x) = 7$, respecting piece conditions.

Question 8

Context

No calculator

A cafeteria model $R(n)$ gives revenue from n meals. Identify independent and dependent variables and give a reasonable domain if at most 480 meals can be prepared.

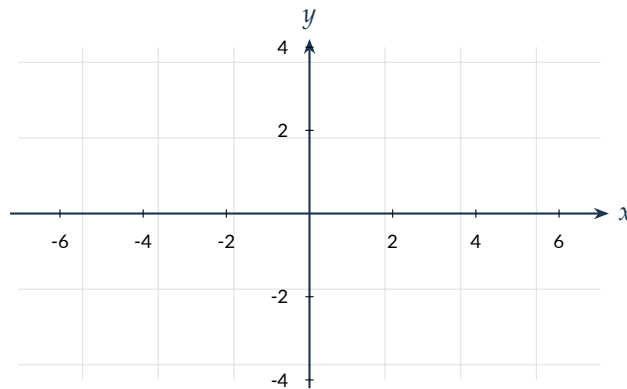
Section C: Challenge and error analysis

Question 9

Challenge

No calculator

Construct two continuous functions on $[0, 4]$ through $(0, 1), (2, 5), (4, 1)$, one with one local maximum and another with at least two.



Question 10

Error analysis

No calculator

A student says, "Since $f(-2) = -7 < 0$, the function is decreasing at $x = -2$." Diagnose the error.

Question 11

Table reasoning

No calculator

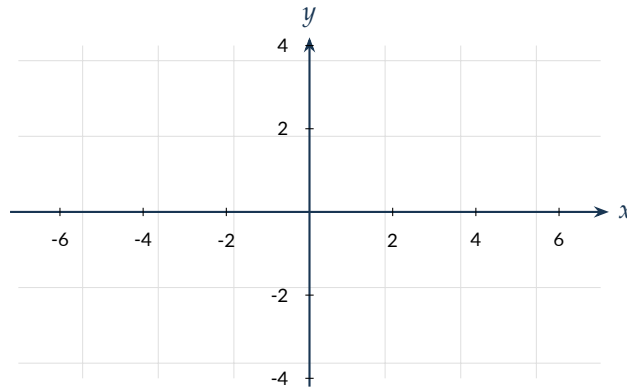
The table gives $m(0) = 5, m(1) = 2, m(2) = -1, m(3) = -1, m(4) = 4$. Describe displayed direction and sign and identify displayed zeros.

Question 12

Challenge

No calculator

Create a function with domain $[-2, 4]$, range $[1, 7]$, decreasing on $[-2, 1]$, constant on $[1, 2]$, and increasing on $[2, 4]$.



Section D: AP-style multiple choice

MCQ 1

No calculator

Which relation defines y as a function of x ?

- A. $\{(-1, 2), (0, 3), (-1, 4)\}$
- B. $x^2 + y^2 = 9$
- C. $y = \sqrt{x + 2}$ for $x \geq -2$
- D. $x = y^2$ for all real y

MCQ 2

No calculator

If $T(d)$ is travel time in minutes for distance d km, what does $T(6) = 42$ mean?

- A. The route is 42 km after 6 min.
- B. A 6 km route takes 42 min.
- C. Time increases 42 min/km.
- D. A 42 km route takes 6 min.

MCQ 3

No calculator

Which description is logically possible on $[0, 5]$?

- A. Negative with all outputs greater than zero.
- B. Positive and decreasing from 12 to 4.
- C. Increasing with $f(0) > f(5)$.
- D. Constant with two outputs at $x = 3$.

MCQ 4

No calculator

$S(t)$ gives students in school t hours after 6 a.m. Which is a reasonable one-day domain?

- A. All real numbers
- B. All integers
- C. $0 \leq t \leq 12$
- D. $S(t) \geq 0$

AP-style free response 1

Calculator permitted - 6 points

A greenhouse records temperature $F(t)$, in degrees Celsius, t hours after 6 a.m.

t	0	2	5	8	11
$F(t)$	22	29	37	34	26

Also $G(x) = 0.4x + 3$.

A. Find $G(F(2))$ and show the intermediate value.

B. Find every displayed preimage of an output greater than 35.

C. Describe displayed covariation and a possible maximum.

D. State a reasonable domain and one limitation of the table.

AP-style free response 2**No calculator - 6 points**

A reservoir contains 900 m^3 at $t = 0$. It decreases steadily to 540 at $t = 3$, remains unchanged until $t = 5$, then decreases at a faster constant rate and is empty at $t = 8$.

A. Draw a labeled graph of $V(t)$.

B. Write a piecewise linear rule.

C. State range and direction intervals.

D. Explain why $V(t) > 0$ does not imply increasing behavior.

Topic 1.2 - Rates of Change

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Calculate and interpret average rates of change from formulas, tables, graphs, and contexts.
- Compare rates on different intervals and estimate a rate at a point using nearby average rates.
- Simplify difference quotients and explain what the algebra reveals about changing rates.

Section A: Foundation and representation

Question 1

Algebra

No calculator

For $f(x) = 2x^2 - 7x + 4$, calculate the average rate on $[-1, 3]$.

Question 2

Table reasoning

No calculator

For points $(0, 6)$, $(2, 15)$, $(5, 21)$, $(9, 5)$, calculate each consecutive displayed average rate.

Section A continued

Question 3

Context

No calculator

Water use rises from 1.8 million L at 6 a.m. to 3.15 million L at noon. Find and interpret the average rate.

Question 4

Graph reasoning

No calculator

A graph contains $A(-3, 8)$, $B(1, -4)$, $C(6, 11)$. Find secant slopes AB and BC . Does the first slope prove the graph decreases everywhere on $[-3, 1]$?

Question 5

Local estimate

Calculator permitted

Given $p(5.98) = 72.421$ and $p(6.02) = 72.589$, estimate the rate at $x = 6$.

Section B: Application and reasoning

Question 6

Symbolic reasoning

No calculator

For $h(x) = 4x^2 - x + 3$, simplify $\frac{h(x+k)-h(x)}{k}$, $k \neq 0$.

Question 7

Parameter

No calculator

For $F(x) = ax^2 + 5x - 2$, the average rate on $[1, 4]$ is 20. Find a .

Question 8

Comparison

No calculator

Two quantities both increase by 24. One change occurs over input length 3, the other over length 8. Compare rates.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student computes the rate from $(2, 7)$ to $(6, 19)$ as $(19 - 7)/6 = 2$. Diagnose and correct.

Question 10

Challenge

No calculator

Construct a quadratic with $q(0) = 4$, rate 1 on $[0, 2]$, and rate 7 on $[2, 4]$.

Question 11

Piecewise modeling

No calculator

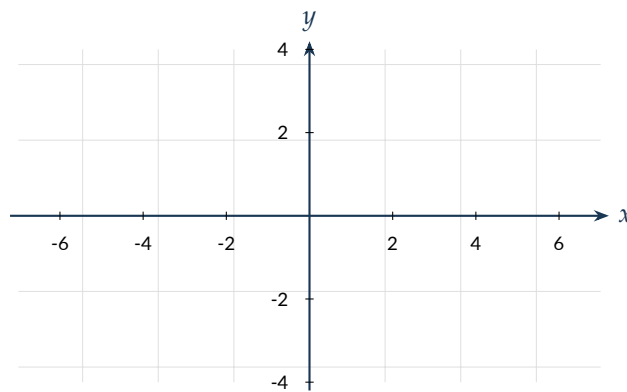
A piecewise-linear graph passes through $(0, 10)$, $(4, 22)$, $(7, 22)$, $(9, 8)$. Write each segment rule and identify greatest rate magnitude.

Question 12

Proof challenge

No calculator

Two continuous functions have the same values at 0, 2, 4. Must they be identical on $[0, 4]$? Relate your answer to average rates with those endpoints.



Section D: AP-style multiple choice

MCQ 1

No calculator

If $f(2) = 11$ and $f(8) = -7$, what is the average rate on $[2, 8]$?

- A. -18
- B. -3
- C. 3
- D. 18

MCQ 2

No calculator

A quantity changes at -4.5 kg/day for 12 days. What is net change?

- A. -54 kg
- B. -16.5 kg
- C. -0.375 kg
- D. 54 kg

MCQ 3

No calculator

What is the average rate of $p(x) = x^2 + 1$ on $[a, a + 3]$?

- A. $2a + 3$
- B. $2a + 6$
- C. $a^2 + 6a + 10$
- D. $3a + 3$

MCQ 4

No calculator

A rate at $x = 5$ is estimated using $[4.99, 5.01]$. Which statement is supported?

- A. It is automatically exact.
- B. It is a nearby secant rate.
- C. It equals the output at 5.
- D. The function must be linear nearby.

AP-style free response 1

Calculator permitted - 6 points

Mass $M(t)$, in grams, is measured at selected times.

t	0	2	5	7	9
$M(t)$	30	52	78	73	45

Also $M(4.8) = 77.6$, $M(5.2) = 78.0$.

A. Calculate and interpret the rate on $[0, 5]$.

B. Calculate the rate on $[5, 9]$ and compare magnitudes.

C. Estimate the rate at $t = 5$.

D. Explain why the data do not prove an absolute maximum at $t = 5$.

AP-style free response 2

No calculator - 6 points

Let $p(x) = ax^2 + bx + c$, $a \neq 0$.

A. Simplify $\frac{p(r+h)-p(r)}{h}$.

B. Show the average rate on $[u, v]$ is $a(u + v) + b$.

C. Rates on $[1, 3]$ and $[3, 5]$ are 4 and 12. Find a, b .

D. If $p(0) = 7$, write p .

Topic 1.3 - Rates of Change in Linear and Quadratic Functions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use first and second differences correctly, including the equal-spacing requirement.
- Construct linear or quadratic models from values, rates, and structural conditions.
- Derive and interpret how average rates of a quadratic change across equal intervals.

Section A: Foundation and representation

Question 1

Foundation

No calculator

For inputs 0, 1, 2, 3, 4, outputs are 8, 13, 18, 23, 28. Classify and write a rule.

Question 2

Finite differences

No calculator

For outputs $-1, 2, 11, 26, 47$ at inputs 0, 1, 2, 3, 4, find a quadratic rule.

Section A continued

Question 3

Missing value

No calculator

A quadratic table for $x = 0, 1, 2, 3$ has outputs $4, 9, 18, k$. Find k .

Question 4

Nonuniform data

No calculator

Inputs $0, 2, 5, 9$ have outputs $1, 7, 28, 82$. Explain why raw differences do not establish degree and compute rates.

Question 5

Construction

No calculator

Determine the quadratic through $(0, -2), (1, 1), (2, 10)$.

Section B: Application and reasoning**Question 6**

Rate structure

No calculator

For $q(x) = 4x^2 - 7x + 2$, find rates on $[1, 3]$ and $[3, 5]$.

Question 7

Inverse reasoning

No calculator

A quadratic has consecutive one-unit rates $-5, -1, 3, 7$. Find its leading coefficient.

Question 8

Parameter

No calculator

For $f(x) = ax^2 + 2x + 1$, rates on $[0, 2]$ and $[2, 4]$ differ by 24. Find a .

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student labels a table quadratic because output differences are 4, 8, 12. State the missing condition and give an input list that invalidates the claim.

Question 10

Challenge

No calculator

Construct a quadratic whose one-unit rate on $[x, x + 1]$ is $6x - 1$ and whose value at 0 is 5.

Question 11

Context

No calculator

A ball has height $H(t) = -5t^2 + 20t + 1$. Find one-second rates on $[0, 1]$, $[1, 2]$, $[2, 3]$ and interpret.

Question 12

Proof challenge

No calculator

Prove that a quadratic expression with zero second difference for nonzero step is actually linear or constant.

Section D: AP-style multiple choice

MCQ 1

No calculator

First differences are $-2, 3, 8, 13$ for unit steps. What is the quadratic leading coefficient?

- A. 2.5
- B. 5
- C. 10
- D. Cannot determine

MCQ 2

No calculator

Which table can be linear?

- A. Inputs 0, 1, 2; outputs 1, 4, 9
- B. Inputs 0, 2, 4; outputs 3, 9, 15
- C. Inputs 0, 1, 3; outputs 2, 6, 10
- D. Inputs 1, 2, 3; outputs 5, 5, 7

MCQ 3

No calculator

For $q(x) = -2x^2 + 7x + 1$, how do consecutive one-unit rates change?

- A. -4
- B. -2
- C. 2
- D. 4

MCQ 4

No calculator

Rates on $[0, 2]$ and $[2, 4]$ are 6 and 14. Find the leading coefficient.

- A. 1
- B. 2
- C. 4
- D. 8

AP-style free response 1

Calculator permitted - 6 points

Production values are

n	0	1	2	3	4
$P(n)$	12	18	28	42	60

A. Use differences to identify a model type.

B. Determine $P(n) = an^2 + bn + c$.

C. Find and interpret the rate on $[1, 4]$.

D. Predict $P(6)$ and state one caution.

AP-style free response 2**No calculator - 6 points**

A quadratic $q = ax^2 + bx + c$ has rate 3 on $[0, 2]$, rate 11 on $[2, 4]$, and $q(1) = 5$.

A. Write two equations from the rates.

B. Find a, b .

C. Find c and write q .

D. Find the second difference for step 3.

Topic 1.4 - Polynomial Functions and Rates of Change

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use finite differences and polynomial structure to identify or justify a likely degree.
- Analyze polynomial rates, turning behavior, zeros, and representations without overclaiming from secant evidence.
- Use technology responsibly to locate features and critique a polynomial model's domain.

Section A: Foundation and representation

Question 1

Finite differences

No calculator

For outputs 2, 4, 18, 56, 130 at $x = 0, 1, 2, 3, 4$, identify the least degree.

Question 2

Missing value

No calculator

A cubic table has outputs 1, 4, 15, 40, k for inputs 0, 1, 2, 3, 4. Find k if third differences are constant.

Section A continued

Question 3

Structure

No calculator

State maximum possible real zeros and turning points for a degree-6 polynomial. Must both maxima occur?

Question 4

Rates

No calculator

For $P(x) = x^3 - 4x$, find average rates on $[-2, 0]$, $[0, 2]$, $[2, 3]$.

Question 5

Local estimate

Calculator permitted

Given $F(2.995) = 17.011$, $F(3.005) = 16.989$, estimate the rate at 3.

Section B: Application and reasoning**Question 6**

Rate pattern

No calculator

One-unit rates over four intervals are $-5, 1, 13, 31$. Identify a likely degree of the original polynomial.

Question 7

Parameter

No calculator

For $P = x^3 + kx^2 - 4x + 2$, the rate on $[0, 2]$ is 6. Find k .

Question 8

Construction

No calculator

Construct a cubic with zeros $-1, 2, 5$ and $P(0) = 20$.

Section C: Challenge and error analysis**Question 9**

Connections

No calculator

Two polynomials share leading term $3x^5$. State one feature they must share and two they need not.

Question 10

Error analysis

No calculator

A cubic appears to have one zero in a calculator window. Explain how other zeros could be hidden and how to verify.

Question 11

Model critique

No calculator

A quartic regression fits five measured points exactly. Give two reasons this is not enough for a ten-year forecast.

Question 12

Challenge

No calculator

Construct a degree-4 polynomial with exactly two distinct real zeros, each of multiplicity 2, and describe its graph at those zeros.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which pattern most strongly indicates a cubic for exact equal-step data?

- A. Constant first differences
- B. Constant second differences
- C. Constant third differences
- D. Alternating first differences

MCQ 2

No calculator

Maximum possible turning points of degree 8?

- A. 7
- B. 8
- C. 9
- D. 16

MCQ 3

No calculator

A polynomial has equal values at $x = 1, 5$. What must be true?

- A. It is constant on $[1, 5]$.
- B. Its average rate on $[1, 5]$ is 0.
- C. It has a zero there.
- D. It is quadratic.

MCQ 4

No calculator

A quartic is fitted on $0 \leq t \leq 6$. Which use is least justified without more evidence?

- A. Evaluate at 3
- B. Rate on $[2, 4]$
- C. Describe a fitted turn in the interval
- D. Predict at 60

AP-style free response 1**Calculator permitted - 6 points**

On $0 \leq t \leq 10$, $H(t) = -0.04t^4 + 0.8t^3 - 4.5t^2 + 6t + 28$.

A. Find local extrema to three decimals.

B. Determine absolute extrema on the domain.

C. Calculate the average rate on $[2, 8]$.

D. Give one reason not to use the model for all $t > 10$.

AP-style free response 2**No calculator - 6 points**

Let $P(x) = (x + 2)(x - 1)^2(x - 4)$.

A. State degree, leading coefficient, and end behavior.

B. List zeros, multiplicities, and cross/touch behavior.

C. Determine sign on intervals formed by zeros.

D. Find the rate on $[0, 2]$ and interpret equal endpoints.

Topic 1.5 - Polynomial Functions and Complex Zeros

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Apply the Fundamental Theorem of Algebra and conjugate-pair theorem to count zeros.
- Construct real-coefficient polynomials from real and complex zeros.
- Connect real-zero multiplicity to graph behavior while distinguishing complex zeros from intercepts.

Section A: Foundation and representation

Question 1

Counting

No calculator

A real degree-6 polynomial has zeros $2 + i$, $-3i$, and 5. List required conjugates and the number of remaining zeros.

Question 2

Multiplicity

No calculator

A real degree-8 polynomial has zeros 1 multiplicity 3, $-2 + 4i$, and $5i$. How many zeros remain after conjugates?

Section A continued

Question 3

Construction

No calculator

Construct the monic cubic with zeros -4 and $2 \pm i$.

Question 4

Complex factoring

No calculator

Factor $x^4 + 5x^2 + 4$ completely over the complex numbers.

Question 5

Graph behavior

No calculator

Describe the real graph behavior at every real zero of $(x + 2)^3(x - 1)^2(x^2 + 9)$.

Section B: Application and reasoning**Question 6**

Parameter

No calculator

If $P(x) = x^3 + kx^2 + x - 2$ has real coefficients and i is a zero, determine k and all zeros.

Question 7

Reasoning

No calculator

Show that $x^2 - 4x + 13$ has no real zeros and state its complex zeros.

Question 8

Degree

No calculator

Find the least possible degree of a real-coefficient polynomial having zeros 2, $1 + i$, and $-3i$.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student marks $2 + i$ as an x -intercept. Diagnose the error and explain how that zero influences a real polynomial.

Question 10

Challenge

No calculator

Construct a quartic with real coefficients, zeros $1 \pm i$, $2 \pm 3i$, and leading coefficient -1 .

Question 11

Vieta

No calculator

For $P = x^3 + x^2 - 7x - 15$, use Vieta's relationships to state the sum and product of all zeros and verify with $3, -2 \pm i$.

Question 12

Challenge

No calculator

Give a degree-5 real polynomial with exactly one distinct real zero and four nonreal zeros. State multiplicities.

Section D: AP-style multiple choice

MCQ 1

No calculator

A real polynomial has zero $3 - 2i$. Which must also be a zero?

- A. $-3 + 2i$
- B. $3 + 2i$
- C. $-3 - 2i$
- D. $2 + 3i$

MCQ 2

No calculator

Which quadratic has zeros $-1 \pm 4i$?

- A. $x^2 - 2x + 17$
- B. $x^2 + 2x + 17$
- C. $x^2 + 2x - 15$
- D. $x^2 - 2x - 15$

MCQ 3

No calculator

A degree-7 real polynomial has three nonreal conjugate pairs. What must be true?

- A. It has no real zeros.
- B. It has at least one real zero.
- C. All real zeros have even multiplicity.
- D. It has exactly seven intercepts.

MCQ 4

No calculator

At a real zero of multiplicity 4, a polynomial graph generally

- A. crosses and changes sign
- B. touches and keeps sign
- C. has a vertical asymptote
- D. has a hole

AP-style free response 1**No calculator - 6 points**

A monic degree-5 polynomial with real coefficients has zeros -2 , $1 + 2i$, and 3 with multiplicity 2.

A. List all zeros with multiplicity.

B. Write factored form over the reals.

C. Expand the polynomial.

D. Describe graph behavior at its real zeros and end behavior.

AP-style free response 2**Calculator permitted - 6 points**

Consider $Q(x) = x^5 - 6x^4 + 11x^3 + 16x^2 - 50x + 52$.

A. Use technology to approximate all zeros.

B. Group the zeros into real zeros and conjugate pairs.

C. Write an exact factored form over the reals suggested by the approximations.

D. Explain why the graph has only one x -intercept.

Topic 1.6 - Polynomial Functions and End Behavior

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine end behavior from degree parity and leading coefficient.
- Express end behavior with words, arrows, and limit notation and justify leading-term dominance.
- Analyze parameter families in which the degree and end behavior change through cancellation.

Section A: Foundation and representation

Question 1

Foundation

No calculator

State both end-behavior limits of $f(x) = 6x^7 - 2x^4 + 100$.

Question 2

Parity

No calculator

State both end-behavior limits of $g(x) = -0.4x^{10} + 90x^9$.

Section A continued

Question 3

Inverse reasoning

No calculator

A polynomial falls on both ends. What can be concluded about degree parity and leading coefficient?

Question 4

Inverse reasoning

No calculator

A polynomial rises left and falls right. What can be concluded?

Question 5

Dominance

No calculator

Justify the end behavior of $x^6 - 10^8x^5$ algebraically, not from coefficient size.

Section B: Application and reasoning**Question 6**

Parameter

No calculator

For $P_k = (k + 4)x^4 - (k - 1)x^3 + 2$, classify $k = -4$ and $k \neq -4$.

Question 7

Parameter

No calculator

Find a so $(a - 2)x^8 + (3a + 1)x^7 + x$ has opposite end directions.

Question 8

Structure

No calculator

Can a degree-6 polynomial with positive leading coefficient have four local maxima and three local minima? Explain.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A graph appears down/down in $[-10, 10]$ for $P = x^4 - 10000x^2$. Diagnose the conflict with algebra.

Question 10

Construction

No calculator

Construct a degree-5 polynomial with right end down, zeros $-1, 2$, and $P(0) = 8$.

Question 11

Context

No calculator

A profit model is valid only for $0 \leq x \leq 40$ but has $P(x) \rightarrow \infty$ as $x \rightarrow \infty$. Explain why the limit need not predict actual unlimited profit.

Question 12

Challenge

No calculator

Determine all k for which $(k^2 - 1)x^6 + (k - 1)x^5 - x^4$ has both ends down.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which end behavior matches $-7x^{12} + x^{20}$?

- A. up/up
- B. down/down
- C. down/up
- D. up/down

MCQ 2

No calculator

A polynomial has left end up and right end down. Which leading term is possible?

- A. $4x^8$
- B. $-4x^8$
- C. $4x^9$
- D. $-4x^9$

MCQ 3

No calculator

For $P_a = (a - 1)x^4 + 3x^3$, when do the ends move oppositely?

- A. $a < 1$
- B. $a = 1$
- C. $a > 1$
- D. All a

MCQ 4

No calculator

Which statement is always true?

- A. A degree-6 polynomial has six real zeros.
- B. A positive even-degree polynomial is increasing everywhere.
- C. Lower-degree terms do not change polynomial end behavior.
- D. An odd-degree polynomial has two upward ends.

AP-style free response 1**No calculator - 6 points**

Let $P_a(x) = (a - 3)x^6 + (2a + 1)x^5 - 7x + 4$.

A. Classify end behavior for $a > 3$.

B. Classify end behavior for $a < 3$.

C. Determine end behavior for $a = 3$.

D. Find all a for which ends move oppositely and justify.

AP-style free response 2**Calculator permitted - 6 points**

Profit, in thousands of QAR, is modeled on $0 \leq t \leq 8$ by $R(t) = -0.5t^4 + 6t^3 - 18t^2 + 40$.

A. Use technology to find all local extrema on the domain.

B. Determine absolute maximum and minimum on the domain.

C. State the polynomial's global end behavior.

D. Explain why global end behavior does not describe profit beyond the stated domain.

Topic 1.7 - Rational Functions and End Behavior

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine rational end behavior by comparing numerator and denominator degrees.
- Use polynomial division to identify slant or polynomial asymptotes.
- Analyze approach from above or below and interpret long-run behavior in context.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find the horizontal asymptote of $\frac{5x^3-2x+1}{-2x^3+7}$.

Question 2

Degree comparison

No calculator

Find the horizontal asymptote of $\frac{4x^2+1}{x^5-3}$.

Section A continued

Question 3

Reasoning

No calculator

Explain why $\frac{x^4+1}{2x^2-5}$ does not have a horizontal or slant asymptote.

Question 4

Division

No calculator

Divide $\frac{3x^2-5x+7}{x+1}$ and state the slant asymptote.

Question 5

Division

No calculator

Find the polynomial asymptote of $\frac{x^4+2x^3-x+1}{x^2+1}$.

Section B: Application and reasoning**Question 6**

Approach

No calculator

For $R = 4 - \frac{7}{x^2+2}$, describe approach to the asymptote.

Question 7

Approach

No calculator

For $S = -3 + \frac{x}{x^2+9}$, describe approach on each end.

Question 8

Construction

No calculator

Construct a rational function with horizontal asymptote $y = 6$, no real vertical asymptotes, and $R(0) = 4$.

Section C: Challenge and error analysis

Question 9

Error analysis

No calculator

A student says $\frac{2x^3+1}{x^2+5}$ has horizontal asymptote $y = 2$ because of leading coefficients. Diagnose.

Question 10

Parameter

No calculator

Find k so $\frac{kx^2+3x}{2x^2-1}$ has horizontal asymptote $y = -4$.

Question 11

Context

No calculator

Average cost is $C(n) = \frac{1200+8n}{n+15}$. Find its long-run value and interpret.

Question 12

Challenge

No calculator

Construct a rational function with asymptote $y = x^2 - 3$, lying above it for all real x in its domain and having no real exclusions.

Section D: AP-style multiple choice

MCQ 1

No calculator

What is the horizontal asymptote of $\frac{-6x^4+x}{3x^4+2}$?

- A. $y = -6$
- B. $y = -2$
- C. $y = 0$
- D. No horizontal asymptote

MCQ 2

No calculator

Which function has slant asymptote $y = 2x + 1$?

- A. $2x + 1 + \frac{3}{x^2+1}$
- B. $\frac{2x+1}{x^2+1}$
- C. $2 + \frac{x}{x^2+1}$
- D. $\frac{x^2}{2x+1}$

MCQ 3

No calculator

If numerator degree is two less than denominator degree, the horizontal asymptote is

- A. the leading ratio
- B. $y = 0$
- C. a slant line
- D. a quadratic

MCQ 4

No calculator

For $R = 5 + \frac{-3x}{x^2+1}$, how does R approach 5 as $x \rightarrow \infty$?

- A. from above
- B. from below
- C. does not approach
- D. alternates

AP-style free response 1**No calculator - 6 points**

Let $R(x) = \frac{2x^3 - x^2 + 5}{x^2 - 4}$.

A. Write quotient-remainder form.

B. State the polynomial asymptote and domain exclusions.

C. Determine approach relative to the asymptote as $x \rightarrow \infty$.

D. Determine approach relative to the asymptote as $x \rightarrow -\infty$.

AP-style free response 2

Calculator permitted - 6 points

Average cost, in QAR per item, is $C(n) = \frac{1200+8n}{n+15}$ for positive integer production n .

A. Find and interpret the horizontal asymptote.

B. Calculate $C(100)$ to three decimals.

C. Solve $C(n) = 12$ and interpret the solution.

D. Explain why the model should not be evaluated at $n = -15$.

Topic 1.8 - Rational Functions and Zeros

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine rational-function zeros while preserving original domain restrictions.
- Use multiplicity and sign charts to describe behavior near zeros.
- Construct rational functions from specified zeros, holes, asymptotes, and scale conditions.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find zeros and exclusions of $\frac{(x-5)(x+2)}{(x+2)(x-1)}$.

Question 2

Factoring

No calculator

Find zeros, exclusions, and multiplicities of $\frac{(x+1)^3(x-4)^2}{(x-2)(x^2+1)}$.

Section A continued

Question 3

Classification

No calculator

For $R = \frac{x^2-4}{x^2-5x+6}$, distinguish zeros, holes, and vertical asymptotes.

Question 4

Multiplicity

No calculator

Determine whether the graph crosses or touches at each zero of $\frac{(x-1)^4(x+3)}{(x+5)^2}$.

Question 5

Sign analysis

No calculator

Create a sign chart for $\frac{(x+2)(x-1)}{x-4}$.

Section B: Application and reasoning**Question 6**

Parameter

No calculator

Find k if $R = k \frac{(x-2)(x+3)^2}{(x-1)(x+4)}$ and $R(0) = 9$.

Question 7

Construction

No calculator

Construct a rational function with only one real zero 3, of multiplicity 2, and one vertical asymptote -1 .

Question 8

Error analysis

No calculator

A student cancels $x - 2$ and then calls $x = 2$ a zero of the simplified numerator. Explain the correct classification.

Section C: Challenge and error analysis**Question 9**

Reasoning

No calculator

A rational function has zero 1 of multiplicity 3 and no vertical asymptote at 1. What sign behavior occurs there?

Question 10

Challenge

No calculator

Construct a rational function with zeros -2 double and 4 simple, hole at 0 , and vertical asymptote 3 .

Question 11

Context

No calculator

For $P(t) = \frac{(t-2)(t-8)}{(t+1)(t^2+4)}$ on $0 \leq t \leq 10$, identify zeros and sign intervals.

Question 12

Logic challenge

No calculator

Is it possible for a degree-3 numerator with real coefficients to have exactly two distinct nonreal zeros and no real zero? Justify.

Section D: AP-style multiple choice

MCQ 1

No calculator

What is the zero of $\frac{(x-3)(x+5)}{(x-3)(x-1)}$?

- A. -5 only
- B. 3 only
- C. -5 and 3
- D. 1 only

MCQ 2

No calculator

At a numerator factor $(x + 2)^4$, with denominator nonzero at -2 , the graph

- A. crosses
- B. touches without sign change
- C. has a hole
- D. has a vertical asymptote

MCQ 3

No calculator

Which factor can add degree without adding real zeros?

- A. $x - 1$
- B. $x^2 - 1$
- C. $x^2 + 1$
- D. $x^2 - 4x + 3$

MCQ 4

No calculator

A canceled factor $(x - a)$ means $x = a$ is

- A. automatically a zero
- B. still excluded
- C. always a vertical asymptote
- D. in the domain after cancellation

AP-style free response 1**No calculator - 6 points**

Let $F(x) = \frac{(x-4)^2(x+1)}{(x-2)(x+1)(x^2+1)}$.

A. State domain exclusions and simplify.

B. Identify zeros and multiplicities.

C. Identify holes and vertical asymptotes.

D. Determine sign intervals and behavior at the zero.

AP-style free response 2**Calculator permitted - 6 points**

On $0 \leq t \leq 10$, $P(t) = \frac{(t-2)(t-8)}{(t+1)(t^2+4)}$ models a signed performance index.

A. Find all zeros and interpret them as threshold times.

B. Determine intervals where the index is positive and negative.

C. Calculate $P(5)$ to three decimals.

D. Explain why the denominator creates no domain problem on the contextual interval.

Topic 1.9 - Rational Functions and Vertical Asymptotes

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Identify vertical asymptotes after cancellation and distinguish them from holes.
- Determine one-sided infinite behavior using sign and multiplicity.
- Construct and interpret rational functions with specified asymptotic behavior.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find holes and vertical asymptotes of $\frac{(x-2)(x+1)}{(x-2)(x-5)}$.

Question 2

Factoring

No calculator

Find vertical asymptotes of $\frac{x^2+1}{(x-3)^2(x+4)}$.

Section A continued

Question 3

One-sided limits

No calculator

Determine one-sided behavior of $\frac{2}{x+1}$ at -1 .

Question 4

Multiplicity

No calculator

Determine one-sided behavior of $\frac{-3}{(x-4)^2}$ at 4 .

Question 5

Sign analysis

No calculator

Analyze $\frac{x+5}{(x-1)^3}$ at 1 .

Section B: Application and reasoning**Question 6**

Sign analysis

No calculator

Analyze $\frac{x-2}{(x+3)^2}$ at -3 .

Question 7

Multiple factors

No calculator

For $\frac{(x-1)(x+2)}{(x+2)^2(x-4)}$, classify $-2, 4$ and analyze one-sided behavior.

Question 8

Construction

No calculator

Construct a function with vertical asymptote 2 , both sides $+\infty$, and zero -1 .

Section C: Challenge and error analysis

Question 9

Error analysis

No calculator

A student says every denominator zero is a vertical asymptote. Give a counterexample and explain.

Question 10

Parameter

No calculator

Find k so $\frac{k}{(x+2)^2}$ approaches $-\infty$ on both sides of -2 .

Question 11

Context

No calculator

For $D(t) = \frac{500}{t-6} + 80$ with $t > 6$, describe behavior near the domain boundary and as $t \rightarrow \infty$.

Question 12

Challenge

No calculator

Construct a function with VAs at -1 and 2 , with both sides down at -1 and opposite signs left down/right up at 2 .

Section D: AP-style multiple choice

MCQ 1

No calculator

For $\frac{x-1}{(x+2)^2}$ near $x = -2$, both sides approach

- A. $+\infty$
- B. $-\infty$
- C. opposite infinities
- D. a finite hole

MCQ 2

No calculator

Which factor creates a hole at $x = 3$ when present in both numerator and denominator?

- A. $x + 3$
- B. $x - 3$
- C. $x^2 + 3$
- D. $3x - 1$

MCQ 3

No calculator

An uncanceled denominator factor $(x - a)^3$

- A. keeps the same sign on both sides
- B. changes sign across a
- C. creates a hole
- D. creates a zero

MCQ 4

No calculator

For $\frac{1}{x-5}$, $\lim_{x \rightarrow 5^-}$ equals

- A. $+\infty$
- B. $-\infty$
- C. 0
- D. 5

AP-style free response 1**No calculator - 6 points**

Let $R(x) = \frac{(x+4)(x-1)}{(x+2)^2(x-3)}$.

A. Identify zeros and vertical asymptotes.

B. Find both one-sided limits at -2 .

C. Find both one-sided limits at 3 .

D. Create a sign chart for all intervals.

AP-style free response 2

Calculator permitted - 6 points

For $t > 6$, a demand index is $D(t) = \frac{500}{t-6} + 80$.

A. Describe behavior as $t \rightarrow 6^+$.

B. Find and interpret long-run behavior.

C. Solve $D(t) = 180$.

D. Explain the contextual domain restriction and why $t = 6$ is not an output event.

Topic 1.10 - Rational Functions and Holes

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Identify holes from canceled factors and determine exact coordinates.
- Distinguish holes from zeros and vertical asymptotes while preserving the original domain.
- Define repaired piecewise functions and construct rational functions with specified holes.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Find the hole and vertical asymptote of $\frac{(x-2)(x+5)}{(x-2)(x-4)}$.

Question 2

Factoring

No calculator

Find all features of $\frac{x^2-1}{x^2+3x+2}$.

Section A continued

Question 3

Coordinate

No calculator

Determine the hole coordinate of $\frac{(x-3)^2(x+1)}{(x-3)(x+4)}$.

Question 4

Reasoning

No calculator

Explain why $\frac{(x+2)(x-1)}{(x+2)^2}$ has no hole at -2 .

Question 5

Repair

No calculator

Repair $\frac{(x-5)(x+1)}{(x-5)(x+3)}$ at its removable discontinuity.

Section B: Application and reasoning**Question 6**

Construction

No calculator

Construct a function with hole $(0, 2)$ and vertical asymptote 3.

Question 7

Construction

No calculator

Construct a function with hole $(2, 5)$, zero 4, and vertical asymptote -1 .

Question 8

Error analysis

No calculator

A student says the simplified rule and original function are equal at the canceled input. Correct the statement.

Section C: Challenge and error analysis**Question 9**

Conceptual

No calculator

Can a vertical asymptote be repaired by assigning a single output? Explain.

Question 10

Multiple holes

No calculator

Find holes and the vertical asymptote of $\frac{(x-2)(x^2-1)}{(x-2)(x-1)(x+4)}$.

Question 11

Challenge

No calculator

Construct a rational function with two holes, no vertical asymptotes, and simplified rule $x^2 + 1$.

Question 12

Proof challenge

No calculator

Prove that multiplying a rational rule by $(x - a)/(x - a)$ can change the function even though the factor equals 1 wherever defined.

Section D: AP-style multiple choice

MCQ 1

No calculator

The hole of $\frac{(x-2)(x+3)}{(x-2)(x+1)}$ is

- A. (2, 0)
- B. (2, 5/3)
- C. (-1, 0)
- D. vertical asymptote $x = 2$

MCQ 2

No calculator

After a common factor cancels, its zero is

- A. restored to the domain
- B. a removable exclusion
- C. always a zero
- D. always a vertical asymptote

MCQ 3

No calculator

Which value repairs $\frac{x^2-4}{x-2}$ at $x = 2$?

- A. 0
- B. 2
- C. 4
- D. undefined

MCQ 4

No calculator

If cancellation leaves another copy of the same factor in the denominator, the input is generally

- A. a hole
- B. a zero
- C. a vertical asymptote
- D. in the domain

AP-style free response 1**No calculator - 6 points**

Let $R(x) = \frac{(x-1)(x+2)^2}{(x-1)(x+2)(x-3)}$.

A. State exclusions and simplify.

B. Find all hole coordinates.

C. Identify vertical asymptotes and zeros.

D. Define a repaired piecewise function at every hole.

AP-style free response 2**Calculator permitted - 6 points**

Let $M(x) = \frac{(x-2)(x^2-1)}{(x-2)(x-1)(x+4)}$.

A. Use algebra to find all exclusions and simplify.

B. Determine both hole coordinates and the vertical asymptote.

C. Use a graph/table to verify the finite values approached at the holes.

D. Explain why graphing only the simplified expression gives an incomplete representation.

Topic 1.11 - Equivalent Representations of Polynomial and Rational Expressions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Rewrite polynomial expressions in factored, expanded, and quotient-remainder forms and explain what each form reveals.
- Use polynomial division and the Remainder and Factor Theorems to make and justify conclusions.
- Rewrite rational expressions while preserving exclusions, holes, zeros, and asymptotes.

Section A: Foundation and representation

Question 1

Factoring

No calculator

Factor $x^3 + 2x^2 - 9x - 18$ completely and identify each zero.

Question 2

Expansion

No calculator

Expand $(x - 4)^2(x + 1)$ and state the degree, leading coefficient, and y -intercept.

Section A continued

Question 3

Representation

No calculator

For $P(x) = -2(x + 1)^3(x - 5)^2$, state two features visible immediately and one feature not immediately visible.

Question 4

Division

No calculator

Divide $3x^3 + x^2 - 10x + 8$ by $x + 2$ and state the quotient-remainder identity.

Question 5

Parameter

No calculator

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 9x - 9$.

Section B: Application and reasoning**Question 6**

Structure

No calculator

A polynomial satisfies $P(x) = (x + 4)(2x^2 - x + 3) - 11$. Find the quotient and remainder when divided by $x + 4$, and evaluate $P(-4)$.

Question 7

Rational structure

No calculator

Simplify $\frac{x^2+x-6}{x^2-4x+3}$ and identify all exclusions, holes, zeros, and vertical asymptotes.

Question 8

Error analysis

No calculator

A student writes $\frac{x^2-9}{x-3} = x + 3$ for all real x . Diagnose and repair the statement.

Section C: Challenge and error analysis

Question 9

Precision

No calculator

Do $f(x) = \frac{x^2-1}{x-1}$ and $g(x) = x + 1$ define the same function if no domain is specified? Justify.

Question 10

Challenge

No calculator

Expand $(3x + 2)^4$ using the Binomial Theorem.

Question 11

Challenge

No calculator

Find a and b so that $x^2 - 1$ is a factor of $P(x) = x^4 + ax^3 + bx^2 + ax + 1$.

Question 12

Modeling

No calculator

A model is written both as $A(t) = 120t - 300 + \frac{1500}{t+5}$ and $A(t) = \frac{120t^2 + 300t}{t+5}$. Explain which form better reveals long-run behavior and which better reveals the original denominator restriction.

Section D: AP-style multiple choice

MCQ 1

No calculator

What is the remainder when $P(x) = 2x^3 - x + 5$ is divided by $x - 2$?

- A. 3
- B. 11
- C. 19
- D. 23

MCQ 2

No calculator

Which expression is equivalent to $\frac{x^2-4}{x^2-x-2}$ on its original domain?

- A. $\frac{x+2}{x+1}, x \neq 2, -1$
- B. $\frac{x+2}{x+1}, x \neq -2, 1$
- C. $\frac{x-2}{x-1}$
- D. 1

MCQ 3

No calculator

If $P(c) = 0$, which statement must be true?

- A. $x + c$ is a factor
- B. $x - c$ is a factor
- C. c is the leading coefficient
- D. The degree is c

MCQ 4

No calculator

Which form most directly displays a polynomial zero of multiplicity 4 at $x = 3$?

- A. Expanded form
- B. Table form
- C. A factor $(x - 3)^4$
- D. Quotient-remainder form

AP-style free response 1**No calculator - 6 points**

Let $P(x) = x^4 - 5x^3 + 2x^2 + 20x - 24$.

A. Show that $x - 3$ is a factor and divide P by $x - 3$.

B. Factor P completely over the reals.

C. Describe every real zero and its crossing or touching behavior.

D. Simplify $\frac{P(x)}{(x-2)(x+2)}$, preserving restrictions and identifying holes.

AP-style free response 2

Calculator permitted - 6 points

A production model is $A(t) = \frac{120t^2 + 300t}{t+5}$ for $0 \leq t \leq 40$, where A is measured in units per day.

- A. Use polynomial division to write A in quotient-remainder form.

- B. Explain how the new form describes the model's long-run trend within its contextual domain.

- C. Solve $A(t) = 2000$ and report the relevant solution to three decimals.

- D. Explain why $t = -5$ remains important even though it lies outside the contextual domain.

Topic 1.12 - Transformations of Functions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Interpret and construct transformations in the form $g(x) = af(b(x-h)) + k$.
- Map points and key features between parent and transformed functions.
- Determine transformed domains, ranges, intercepts, asymptotes, and holes with justification.

Section A: Foundation and representation

Question 1

Interpretation

No calculator

Describe every transformation from $f(x)$ to $g(x) = -4f(2(x+3)) + 1$.

Question 2

Point mapping

No calculator

If $(8, -3)$ lies on f , find the corresponding point on $g(x) = 5f((x-2)/4) + 7$.

Section A continued

Question 3

Parameter

No calculator

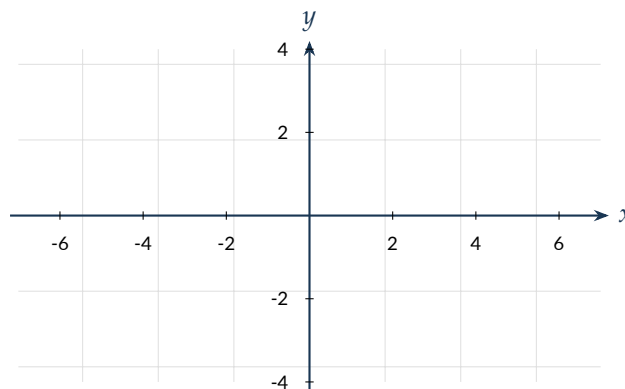
A point $(1, 6)$ on f maps to $(-2, -5)$ under $g(x) = af(3(x + 7/3)) + k$. If $a = -1$, find k .

Question 4

Quadratic

No calculator

Analyze and sketch $p(x) = \frac{1}{2}(x - 4)^2 - 3$: state transformations, vertex, axis, domain, range, and intercept strategy.



Question 5

Absolute value

No calculator

Analyze $v(x) = -2|x + 1| + 6$ and solve $v(x) = 0$.

Section B: Application and reasoning**Question 6**

Root function

No calculator

For $c(x) = \sqrt[3]{-2(x-5)} + 4$, identify the image of the parent origin and state domain and range.

Question 7

Rational

No calculator

Identify all asymptotes, domain, range, and intercepts of $r(x) = \frac{6}{x+2} - 1$.

Question 8

Restrictions

No calculator

The graph of f has a hole at $(2, -1)$. Find the transformed hole for $g(x) = 3f(-2(x-4)) + 5$.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student says $f(3(x - 2))$ means shift right 6 because the inside expands to $3x - 6$. Diagnose the reasoning.

Question 10

Construction

No calculator

Construct a transformed reciprocal function with vertical asymptote $x = 5$, horizontal asymptote $y = -2$, and passing through $(6, 1)$.

Question 11

Parameter

No calculator

Find a and h if $g(x) = a(x - h)^2 + 7$ has vertex $(3, 7)$ and passes through $(5, -1)$.

Question 12

Challenge

No calculator

Given $g(x) = -3f(2x - 8) + 5$, solve for $f(u)$ in terms of g and u .

Section D: AP-style multiple choice

MCQ 1

No calculator

If $(4, 7)$ lies on f , which point lies on $g(x) = 2f(x - 3) - 5$?

- A. $(1, 9)$
- B. $(7, 9)$
- C. $(7, 2)$
- D. $(1, 2)$

MCQ 2

No calculator

The graph of $f(5x)$ is a horizontal

- A. stretch by 5
- B. compression by $1/5$
- C. shift right 5
- D. vertical stretch by 5

MCQ 3

No calculator

What is the horizontal asymptote of $-4/(x + 7) + 3$?

- A. $x = -7$
- B. $x = 3$
- C. $y = -7$
- D. $y = 3$

MCQ 4

No calculator

If the range of f is $[1, 6]$, what is the range of $g(x) = -2f(x) + 5$?

- A. $[-7, 3]$
- B. $[3, 17]$
- C. $[-12, -2]$
- D. $[-17, -3]$

AP-style free response 1

No calculator - 6 points

The table gives selected points on f .

x	-4	-1	2	5
$f(x)$	3	-2	5	1

Define $g(x) = -2f(3(x - 1)) + 4$.

A. Create the corresponding table of four points on g .

B. If the domain of f is $[-4, 5]$, find the domain of g .

C. If the range of f is $[-2, 5]$, find the range of g .

D. Explain how the sign of the outside scale affects the order of output values.

AP-style free response 2

Calculator permitted - 6 points

A transformed reciprocal model has the form $R(x) = a/(x - h) + k$. Its vertical asymptote is $x = 4$, horizontal asymptote is $y = -2$, and the graph passes through $(6, 3)$.

A. Determine a, h, k and write the model.

B. Find both intercepts, giving decimal approximations if needed.

C. Describe the one-sided behavior at the vertical asymptote.

D. State domain and range and explain why neither asymptote is part of the graph.

Topic 1.13 - Function Model Selection and Assumption Articulation

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use differences, ratios, rates, and graphical shape to compare candidate model families.
- Compute and interpret residuals and use residual patterns to evaluate fit.
- State assumptions, contextual domains, interpolation limits, and extrapolation risks in precise language.

Section A: Foundation and representation

Question 1

Pattern

No calculator

For equally spaced inputs, outputs are 11, 17, 23, 29, 35. Select a model family and justify using differences.

Question 2

Pattern

No calculator

For equally spaced inputs, outputs are 2, 6, 18, 54, 162. Select a model family and identify the growth factor.

Section A continued

Question 3

Construction

No calculator

Outputs are 4, 7, 12, 19, 28 for inputs 0, 1, 2, 3, 4. Select and construct a model.

Question 4

Residual

No calculator

An observed output is 83 and a model prediction is 79.5. Compute and interpret the residual.

Question 5

Residual analysis

No calculator

Model A residuals are $-1.2, -0.8, -0.1, 0.7, 1.4$; Model B residuals are $0.5, -0.4, 0.2, -0.3, 0.1$. Which is more defensible and why?

Section B: Application and reasoning**Question 6**

Error analysis

No calculator

A model has $R^2 = 0.998$ but a U-shaped residual plot. Evaluate the model choice.

Question 7

Assumptions

No calculator

A school attendance model assumes enrollment and transport access remain stable. Explain why each assumption matters.

Question 8

Domain

No calculator

A data set records the number of defects in each manufactured item. Should the input domain be continuous or discrete? Explain.

Section C: Challenge and error analysis**Question 9**

Model selection

No calculator

A population grows by roughly 8% per year for six years. Compare linear and exponential interpretations and select one if the percentage is stable.

Question 10

Challenge

No calculator

A chemical response rises rapidly and then approaches a stable ceiling. Explain why an unrestricted polynomial may be less plausible than a rational model with a horizontal asymptote.

Question 11

Investigation

No calculator

Two models fit sales data from months 1 through 6. At month 6, one has a large positive residual while earlier residuals are near zero. What should be investigated?

Question 12

Critique

No calculator

A student uses a model fitted to temperatures from 15°C to 30°C to predict at 100°C and reports six decimal places. Identify two weaknesses.

Section D: AP-style multiple choice

MCQ 1

No calculator

A residual of -6 means the model

- A. underpredicted by 6
- B. overpredicted by 6
- C. has slope -6
- D. is invalid

MCQ 2

No calculator

Which pattern most strongly supports an exponential model at equally spaced inputs?

- A. Constant first differences
- B. Constant second differences
- C. Constant ratios
- D. Constant residuals equal to 5

MCQ 3

No calculator

Which residual plot is most desirable?

- A. A clear upward line
- B. A U-shaped curve
- C. Small points randomly scattered around zero
- D. All residuals positive

MCQ 4

No calculator

Which statement is an assumption rather than a computed result?

- A. The residual is 2.3
- B. The market conditions remain similar next year
- C. The quadratic coefficient is -4
- D. The table contains eight points

AP-style free response 1**Calculator permitted - 6 points**

A laboratory records the values below.

x	0	1	2	3	4	5
y	4.1	7.0	14.2	24.8	40.1	59.0

Quadratic predictions are 4, 7, 14, 25, 40, 59 and exponential predictions are 4.8, 7.5, 11.7, 18.3, 28.6, 44.7.

A. Compute all residuals for both models.

B. Select a model and justify using residual magnitude and pattern.

C. Use the selected model to predict at $x = 8$.

D. State one assumption and one limitation of that prediction.

AP-style free response 2**No calculator - 6 points**

A quantity has values 80, 48, 28.8, 17.28, 10.368 at years 0, 1, 2, 3, 4.

A. Use numerical structure to select a model family.

B. Construct a model $Q(t)$.

C. Interpret the key parameter controlling change.

D. Evaluate a prediction at $t = 12$ and explain whether it should be trusted automatically.

Topic 1.14 - Function Model Construction and Application

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Construct linear, quadratic, polynomial, rational, and piecewise models from stated conditions or data.
- Use models to solve equations, compare rates, and interpret parameters and outputs in context.
- Validate a model against conditions, domain, residual evidence, assumptions, and practical constraints.

Section A: Foundation and representation

Question 1

Linear model

No calculator

Construct the linear model through $(2, 11)$ and $(7, 31)$ and interpret its slope if x is hours and y is liters.

Question 2

Quadratic

No calculator

Construct the quadratic with vertex $(4, -3)$ through $(1, 15)$.

Section A continued

Question 3

Factored model

No calculator

Construct a quadratic with zeros -2 and 6 and passing through $(0, 24)$.

Question 4

Polynomial

No calculator

Construct the least-degree polynomial with zeros -1 simple and 3 double and with $P(0) = -18$.

Question 5

Rational

No calculator

A rational model has horizontal asymptote $y = 50$, vertical asymptote $x = -4$, and passes through $(0, 70)$. Construct $R(x) = a/(x + 4) + 50$.

Section B: Application and reasoning**Question 6**

Piecewise

No calculator

A service costs 30 riyals for the first 5 units and 4 riyals for each additional unit, up to 20 units. Construct a continuous piecewise model.

Question 7

Application

No calculator

Using $M(t) = 80 + 500/(t + 5)$, solve $M(t) = 100$ and interpret the result if t is years.

Question 8

Rate

No calculator

For the arch $h(x) = -0.4x(x - 10)$, find the average rate of change from $x = 1$ to $x = 4$ with units of meters per horizontal meter.

Section C: Challenge and error analysis**Question 9**

Parameters

No calculator

In $R(t) = L + A/(t + b)$, interpret L and explain the effect of increasing $|A|$ when $t + b$ keeps the same sign.

Question 10

Validation

No calculator

A quadratic regression predicts 12.4, 18.1, 21.7 where observations are 12.2, 18.3, 21.6. Compute residuals and briefly validate the local fit.

Question 11

Challenge

No calculator

Give two different function families that can pass through $(0, 1)$ and $(1, 2)$. Explain why two points do not uniquely determine a general model family.

Question 12

Critique

No calculator

A model predicts -14 customers at a future time. Explain how to respond mathematically and contextually.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which form is most efficient for a quadratic with known zeros 2 and 7?

- A. $a(x - 2)(x - 7)$
- B. $a(x - h)^2 + k$
- C. $mx + b$
- D. $a/(x - h) + k$

MCQ 2

No calculator

A model $M(t) = 60 + 240/(t + 3)$ has long-run value

- A. 3
- B. 60
- C. 80
- D. 240

MCQ 3

No calculator

Which condition is necessary after solving a piecewise equation?

- A. The answer is positive
- B. The answer lies in the interval for the branch used
- C. Every branch has the same slope
- D. The graph is a polynomial

MCQ 4

No calculator

A quadratic has vertex $(2, 5)$ and passes through $(4, 13)$. What is its leading scale?

- A. -2
- B. 1
- C. 2
- D. 4

AP-style free response 1**Calculator permitted - 6 points**

The measured height of an arch is shown below.

x	0	2	4	6	8	10
$h(x)$	0	6.4	9.6	9.6	6.4	0

Distances are in meters.

A. Use quadratic regression or structural reasoning to construct a model.

B. Determine and interpret the maximum height and its location.

C. Find the average rate of change from $x = 2$ to $x = 8$.

D. Explain one limitation of applying the model beyond $0 \leq x \leq 10$.

AP-style free response 2**No calculator - 6 points**

A quantity is modeled by $M(t) = L + A/(t+b)$ for $t \geq 0$. It approaches 80 as t increases, and $M(0) = 180$, $M(5) = 130$.

A. Determine L, A, b and write the model.

B. Solve $M(t) = 100$.

C. Find and interpret the average rate of change from $t = 0$ to $t = 5$.

D. State the algebraic restriction and a contextual assumption needed for future use.
